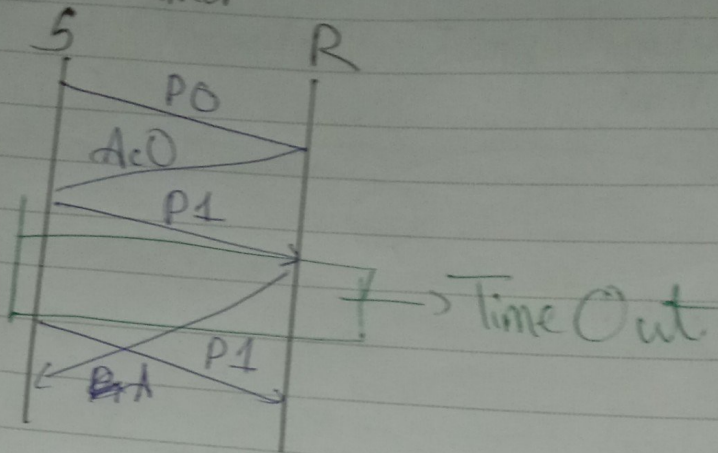


rdt3 Channels with error & loss
We add Timer



- 1) No Loss
- 2) Packet loss
- 3) Acknow loss
- 4) Premature Time Out / Delayed Acknow

~~Final~~ Report

FINAL TERM

17/12/24

(Absent)

3.0 Pipeline

Go back N

Acknowledge

pg 70
sender

||||

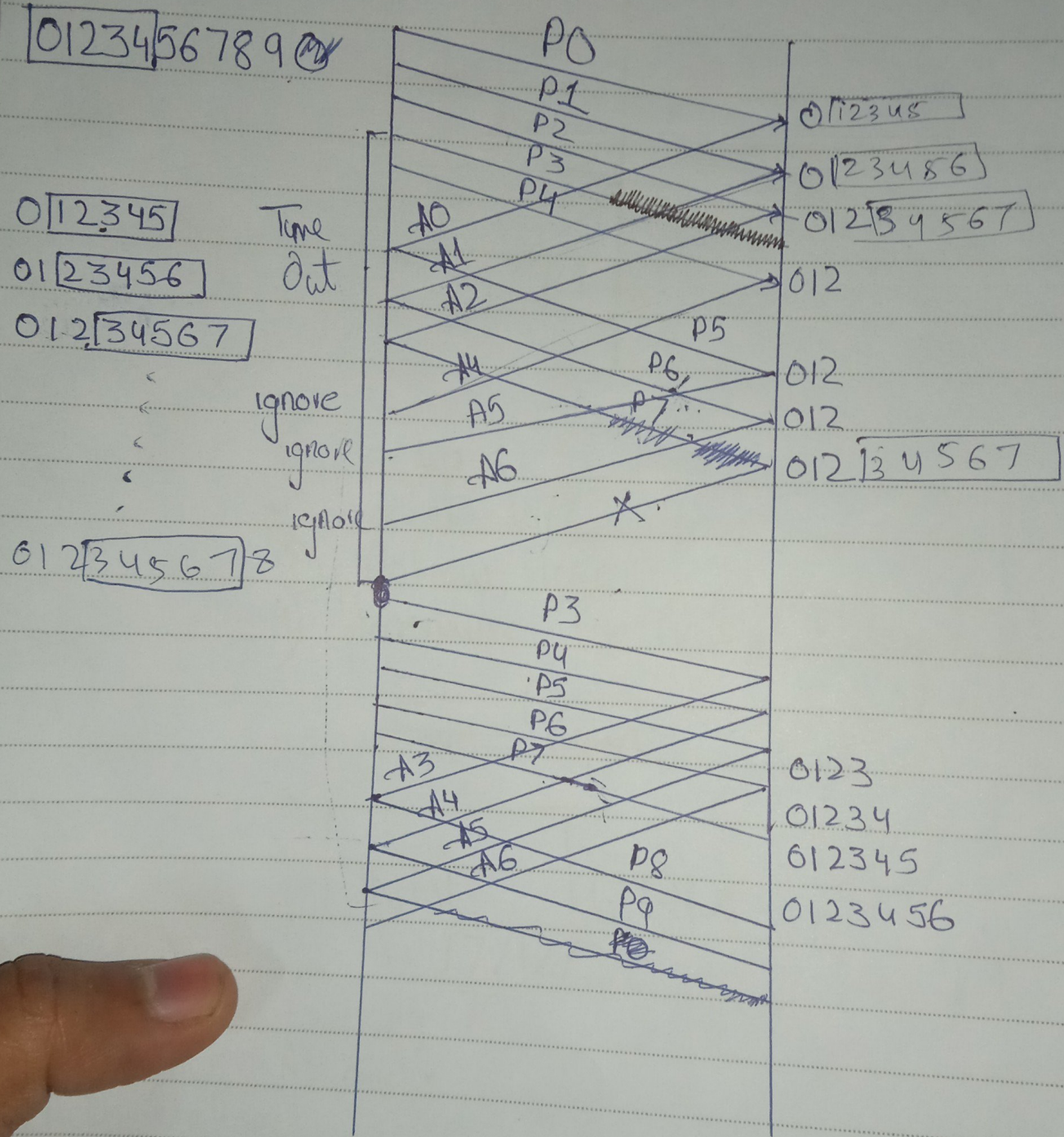
||||

||||

||||

Window
Size

18/12/24
 N=5 P1-3 lost, 7 lost, Total Packet send 10



Not Comp

CHAP 4

Date: _____

24/12/24

* Two key network Layer Functions *

Network Layer

- 1) Data Plane, per router function
- 2) Control Plane, network wide logic
- 3) Per Router Control

4) ~~SDN~~ SDN (Software define Networking)

↳ No dynamic change, software define

5) Network Service Model → questions

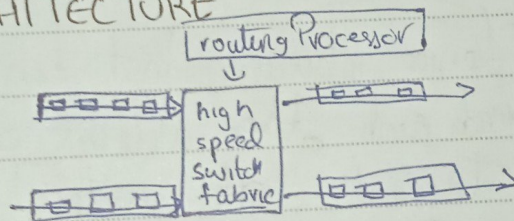
Reflection on best effort service

* 4-12

• Simplicity, sufficient provisioning of bandwidth, replicated application layer distributed service, congestion control of elastic service help.

Routing Algorithm

ROUTER ARCHITECTURE



4-14

Input Port Function
Physical layer

Line Termination

Link layer Protocol

lookup forwarding opening

4-16

line speed → No queue

decentralized forwarding, generalized switching forwarding

Longest Prefix Matching

4-22 →

Ans 1

39-43X

Switching Fabric

input
output

Memory
Bus

Interconnection Network

N times line rate desirable

Input Port Queuing

*(HOL)

Head of Line Blocking

4-30

4-31

* Longest Prefix

33X

4-1 - 4-38

1/1/25

4-43 se start

Network Layer: Internet

Path Select, IP Protocol, ICMP Protocol

IP Address, Subnet (device interface that can physically reach each other without passing through ~~one~~ router.)

* 4-49

1/24

subnet mask

Recipe -> create island
8 -> host

24+8

Akhri bit nhi likhi

223.1.2.6 k subnet ko
222.1.2/24 (subnet part)

CIDR → Classless Inter Domain Routing

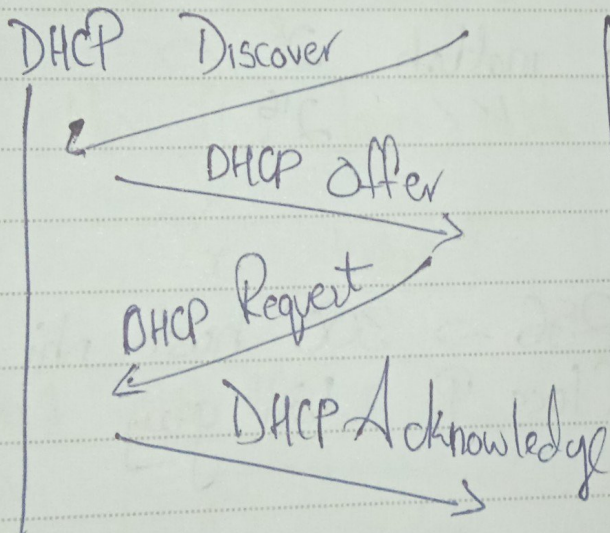
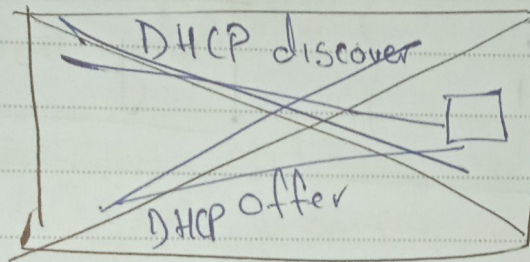
Date: _____

Class A → 255.0.0.0/8 (8)
Class B → 255.255.0.0/16 (16)
Class C → 255.255.255.0/24 (24)
Class D → 255.255.255.0/24

Need ha CIDR se assign krdo.
a.b.c.d/23 → a baaki → 29 → 512 device max left

DHCP → dynamically obtain IP address from network server
↳ router

4-55



IP client ko kaise milega?
IP add ISP ko kaise milega?

60
4-61, 62 X

Date: 2/1/25

NAT Partner ha

NAT Network

2/1/25

4-65

Address Translation

Classfull IP Address
Classless CIDR

Acc. to need

Classfull IP	Address
A	1248
B	1616
C	1824
D	

difference
Classless CIDR

1 x $\rightarrow$ Jitre zarorat.

why using CIDR

+ 24 Subnet * 8 host

8 ka matlab 2^8
16 " " 2^{16}

$2^8 = 256$ unique host

300

$2^8 = 256 \rightarrow 300$ host nhi atay.
 2^{16} Class B kien gay

$2^{16} = 65536$

32 fixed

Classless

Date: 2/1/25

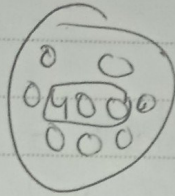
123 $\rightarrow$ subnet $\rightarrow$ 9 left (Host)

$2^9 = 512 \rightarrow$ de sakte hain

There are 3 organization, 1

① Why do we use CIDR

② What is an issue with classful Routing.
(Justify with example)



$2^{10} = 1048$

③ 3 organization, org1 have 1000 devices, org2 have 50,000 devices, org3 have 500 devices
Write CIDR address

Max PC laga sakte ho?

/x kitne?

④ Org1 198.168.0/25 $\rightarrow 32-25=7 \rightarrow 2^7=128$.

Routing

NAT Table

NAT → Network Address Translation #4-68

NAT TABLE

Source IP, Port# / New Port N#, NAT IP Address

IPv6

IPv6 datagram format

Compare IPv4 IPv6 datagram format

No Checksum

No Re assembly

No option

Configurable as upper layer, next header protocol at (router)

Tunneling

4-72 diagram
4-75

Tunneling encapsulate 4-75

4-75 end

77 take read

Routing Protocol 7/11/25

CHAP 5

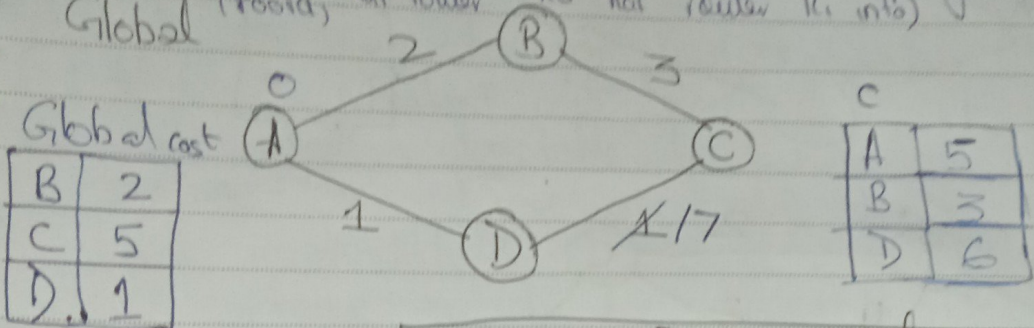
Graph Abstraction → Link Cost

5-10

5-11

Static, Global, Decentralized, Dynamic

Global (Router has router info)



D	
A	1
B	3
C	6

Decentralized (Iterative Process)

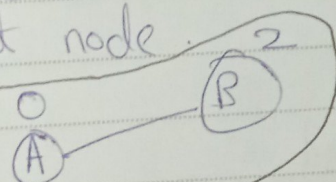
A	
B	2
C	∞
D	1

Link State (Global Information about router)

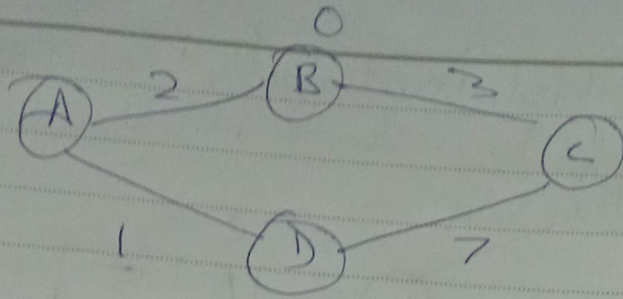
Dijkstra's Link State Routing Algorithm

- ① Centralized $\rightarrow$ network topol, Link cost known to all nodes
- ② Iterative $\rightarrow$ After k iteration, know least cost path to k destination
- ③ Computer Least cost Paths from One Node (Source) to all other nodes. Gives forwarding table for that node.

Source ki cost 0 hoti ha.

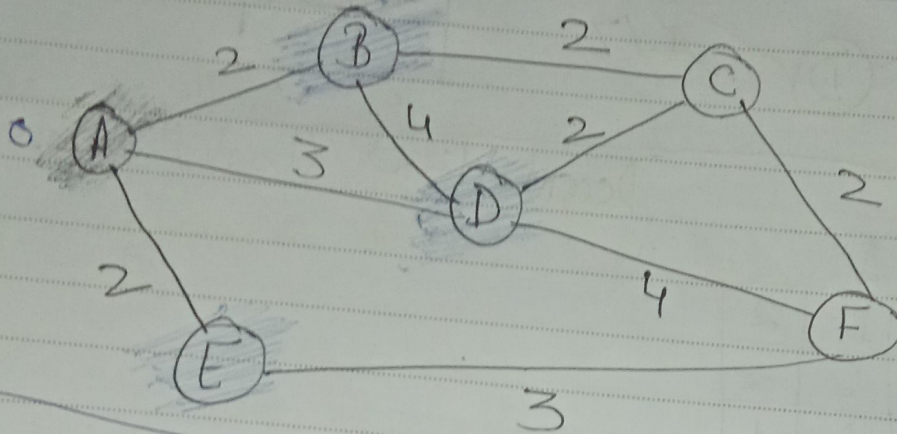


Direct link cost from node x to y; ∞ if not direct neighbors.



	B
A	2
C	3
D	3

Q1.)



○ → nodes

① Directly Connected

A Source Co

Marked

	A	B	C	D	E	F
A	0	∞	∞	∞	∞	∞
B		2	∞	∞	∞	∞
E			4	3	2	∞
D			4	3	2	∞
C			4	3	2	∞
F			4	3	2	∞

6 not better than 3

se Direct
A and B se Direct

Cost

A-A = 0
A-B = 2
A-C = 4
A-D = 3
A-E = 2
A-F = 5

Path

A ←
B ← A
C ← B ← A
D ← A
E ← A
F ← E ← A

- ① A is source
- ② ~~Will~~ ~~no~~ Will mark A
- ③ Will c the directed path from A (B, D, E)
- ④ ~~At~~ Check Shortest cost, mark it.
- ⑤ Path nikalte waqt table ko dekhien gay, sab se last se uper jaye gay, value change, toh hum path mein likh jaye gay. 5 $\rightarrow$ ∞ se change ho raha hai.

Find shortest Path and Cost from A to all other nodes in the network?

D is source now.

Marked	A	B	C	D	E	F
D	∞	∞	∞	0	∞	∞
C	3	4	2		∞	4
	3	4			∞	4

I

★ 8/1/25 ★

Date: 8/1/24

Distance Vector

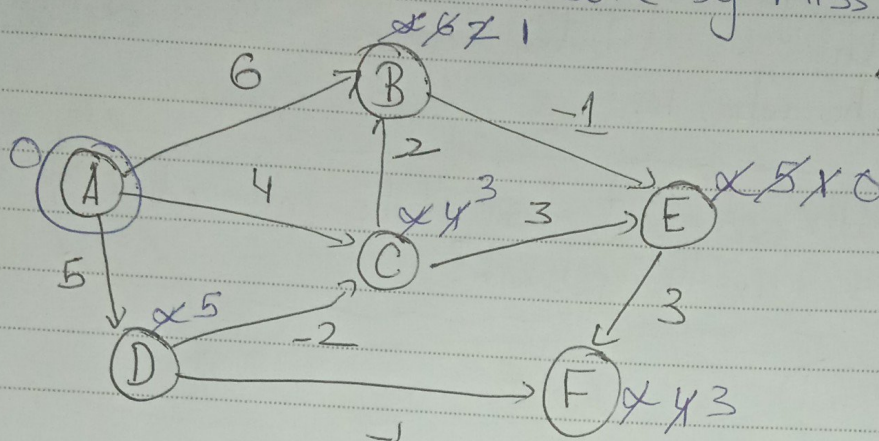
(Sirt Source 0)

5-36

5-44

Neighbors Information in table

Formula Slides ✗ Formula / Solve by Miss ✓



BELLMAN FORD

Vertices →

(A,B) (A,C) (A,D) (B,E) (C,B) (C,E) (D,C) (D,F) (E,F)

$n-1 \Rightarrow 9-1=8$ iterative

1st Iteration, 2nd Iteration, 3rd Iteration

4th Iteration → No change stop, we have find cost.

↳ B, E = 0, F → F = 3

Cost from A

Cost from A

A

B = 1

C = 3

D = 5

E = 0

F = 3

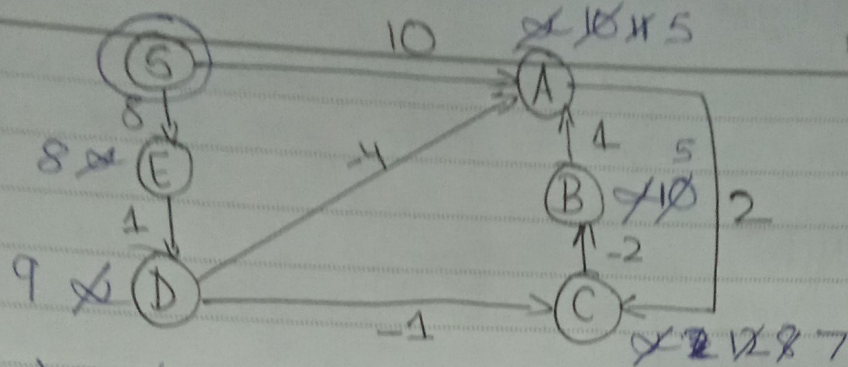
A → B = A → D → C → B

A → C = A → D → C

A → D = A → D

A → E = A → D → C → B → E

A → F = A → D → C → B → E → F



Edges = 8
Iteration = 8 - 1 = 7

	(S,A)	(S,E)	(A,C)	(B,A)	(C,B)	(D,A)	(D,C)	(E,D)
1st Iter	✓	✓	✓	✓	✓	✓	✓	✓
2nd Iter	✓	✓	✓	✓	✓	✓	✓	✓
3rd Iter	✓	✓	✓	✓	✓	✓	✓	✓

Cost from S

$B = A = 5$
 $C = B = 5$
 $D = C = 7$ ✓
 $E = D = 9$
 $A = E = 8$ ✓

Path

$S \rightarrow A = S \rightarrow E \rightarrow D \rightarrow A$ ✓
 $S \rightarrow B = S \rightarrow E \rightarrow D \rightarrow A \rightarrow C \rightarrow B$ ✓
 $S \rightarrow C = S \rightarrow E \rightarrow D \rightarrow A \rightarrow C$
 $S \rightarrow D = S \rightarrow E \rightarrow D$ ✓
 $S \rightarrow E = S \rightarrow E$ ✓

Quiz 3

Subnet

CiDR

Dijkstra

Bellman Ford

Compute
CLO-3

Good

138-C

CHAPTER 6 Link Layer

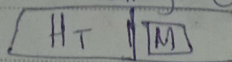
responsibility of

Date: 9/1/25

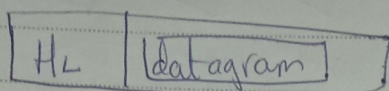
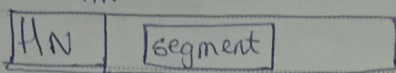
Link Layer → transferring datagram from one node to physically adjacent node over link.

Host & Router → Nodes

2 packet frame, encapsulate datagram



~~H_T~~ → ~~so~~



- datagram transferred by different link protocol over different link.
- each link provide different services.

6-5 Transportation Analogy

Communication Link → Wire / Wireless

- ① Framing Link Access : Need MAC Address in frame header to identify source, destination
- ② Reliable delivery b/w adjacent nodes
- ③ Flow Control
- ④ error detection
- ⑤ Error Correction
- ⑥ Half duplex & Full Duplex

Link Layer implemented in Network Interface Card (NIC) or chip

Multiple Access X

Se Shuru 6-11

Error Detection 6-11
Parity Checking 6-12

Even Parity, Odd Parity 6-12
↳ Only check 1

2-1) Parity → Detect & correct single bit error

1	0	1	0	0
1	1	0	1	1
1	0	1	1	1
1	1	0	0	

1	0	0	0	1
1	1	1	1	0
0	1	1	1	0

Checksum

Cyclic Redundancy Check CRC polynomial

More Powerful Error Detection Coding

$x^3 + x^2 + x^1 + x^0$ → Polynomial

$$x^3 + x^2 + 1$$
$$1 \quad 1 \quad 0 \quad 1$$

↳ x^1 nhi hai.

(+) → Different per 1.

Date: 22, 23, 24

Data → Polynomial Generator

$x^3 + x^2 + 1 \rightarrow$ key

1 0 0 1 0 0

$x^5 + x^2$

1 1 0 1

CRC bits

1101) 1001000000000000
 (+) 1101111111111111
 1000000000000000
 (+) 1101111111111111
 1010000000000000
 (+) 1101111111111111
 1110000000000000
 (+) 1101111111111111
 * 1110000000000000
 (+) 1101111111111111
 0001000000000000

Left most
0 data do
key se
divide

Remainder

1101) 1001000000000000
 (+) 1101111111111111
 x 1000000000000000
 (+) 1101111111111111
 x 1010000000000000
 (+) 1101111111111111
 x 1110000000000000
 (+) 1101111111111111
 x 1401000000000000
 (+) 1101111111111111
 0000000000000000

CRC bit = key - 1

10 Mar → Compute Theory

Date: _____

Data 10110011

$$M = x^7 + x^6 + x^3 + x^2$$

1 1 0 0 1 1 0 0

$$G = 11001$$

$$11001 \overline{) 11001100000000}$$

11001	1100	1100	00	00	00	00	1101
	11001	↓		1	1		01
		10000					
		⊕ 11001					
		× 10011					
		⊕ 11001					
			10100				
			⊕ 11001				
				1101			

CHAP 7

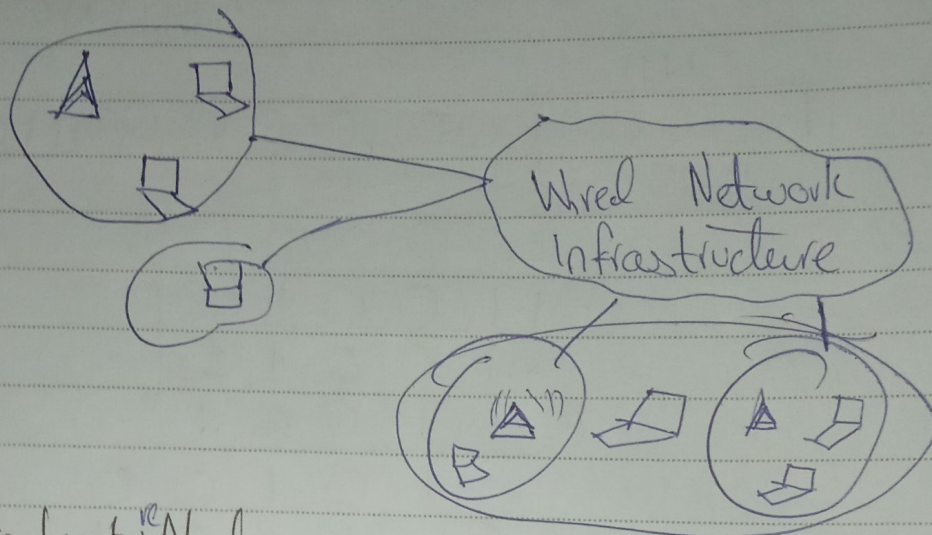
Date: 14/1/25

Wireless and Mobile Network

7-1

Challenges $\rightarrow$ Wireless & Mobility
Wired Network Infrastructure
Wireless does not mean mobility

Base Station



Elements $\rightarrow$ Infrastructure Mode
ad hoc mode

Single, Multiple Hops

Difference b/w Wireless / Wire.

7-15

Characteristic Hidden terminal, Signal Attenuation

7-17* $\rightarrow$ 7-20X

7-21

LAN architecture

Passive / Active Scanning

Hidden / 2

Date: 15/1/25

I am AP who want 2 connect
new device

Infrastructure, Adhoc

Wireless vs Wired

7-15 *

Hidden Problem

X 7-16-721 X

7-23 * Active Passive
Scanning

7-24

802.11 → No Collision
↳ Avoid

7-25

802.11 sender Receiver

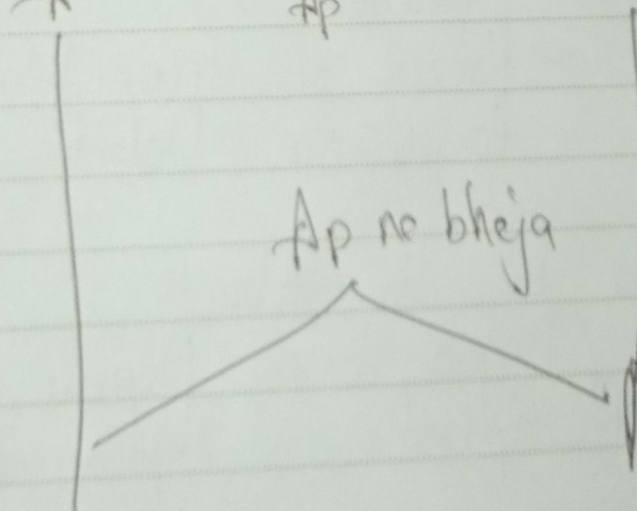
RTS → small Request to send

A

Ap

Ap no bheja

7-27



CLO1 → 10 mark describe (3, 4)
2 → Numerical
3 →

Date:

16/1/25

Security Attack

Active Passive Threat

Passive Attack → Interception
↳ Read
↳ Monitoring

Active Attack → Interrupt
→ Fabrication
→ Replay
→ Modification

Handling Attack
Passive Attack → focus on Prevention
Active Attack → focus on Detection

How to do Attack
Social Engineering, Virus/Trojan Horse, Denial of Service
IP Spoofing, Replay Attack

Date: _____

Denial of Service

IP Spoofing , Replay Attack

Active / Passive

Social Engineering Scenario

3/4 attack

CIO1 → X

3 → Sliding Window Protocol
Reliable
Ackno

Packet

Sequence

CRC, 2D Parity, 2 Routing
is Concept.